BondBreak Strategy Router

Local pipeline runner for Modules -1–3 with clear, auditable routing output.

Pipeline Output

PIPELINE DECISION

NO_GO

FAIL | Mode standard | M2 DEGRADED_OK | M3 PASS | TopK 0

PIPELINE HALTED

M0_NO_MATCH

NEXT INPUT REQUIRED

target_bond

ERRORS

No bonds matched the requested token.

WARNINGS

Token inferred heuristically from input., Requested output appears to be a functional group., W_MODULE1_VETO: downstream physics/geometry indicates reroute or review.

Job Type	Selected Bond	Primary Route
REAGENT_GENERATION	n/a	n/a

Difficulty	Compute Active	Scaffolds
HARD	n/a	n/a

SCIENTIFIC REPORT

PIPELINE HEALTH & INTERCONNECTION

MODULE	STATUS	RESULT	ATTENTION
Module -1	PASS	ch	Warnings in SRE
Module 0	NO_GO	n/a	Needs target clarification
Module 1	FAIL	Top 0 scaffolds	TopoGate failed
Module 2	DEGRADED_OK	n/a	OK
Module 3	PASS	Plan 0.142 / Phys 0.000	OK

Working now

- Module -1 reactivity: PASS (constraint: NONE)
- Negative control assigned

Results snapshot

- Module -1: PASS (constraint: NONE)
- Route: n/a
- Module 1: FAIL
- Module 2: n/a
- Module 3 arms: 5
- Neg control: A4
- Halt reason: M0_NO_MATCH

Needs improvement

- Module 0 halted: M0_NO_MATCH
- Module 1 status: FAIL
- Route confidence low

Why this score

- Route confidence 0.00 below 0.60 (Route confidence below this indicates weak mechanistic support.)
- Target resolution 0.00 below 0.85 (Target resolution below this indicates ambiguous bond selection.)
- Plan phys 0.00, SNR=0.0

Handoff checks

- OK M-1 → M0: resolved_target + selected candidates
- Missing M0 → M1: selected_bond
- Missing M1 → M2: top_scaffolds
- Missing M2 → M3: module3_handoff
- OK Shared IO: condition_profile + telemetry

CONDITION FLOW (SHARED IO)

pH	Temp C	Ionic strength
n/a	n/a	n/a

Solvent	Cofactors	Run ID
n/a	n/a	f1a98533-821d-4a93-b6a2-40abaf5f0334

MODULE	GIVEN EFFECT	OPTIMUM	CONFIDENCE	RETRY
Module 0	n/a	n/a	n/a	n/a
Module 1	n/a	n/a	n/a	insufficient_data
Module 2	n/a	n/a	n/a	unknown_optimum
Module 3	Baseline activity under user conditions.	n/a	0.14	not_applicable

SHARED IO DETAILS

Trace module-1, module1, module2, module3

Bond Spec C-H

EVIDENCE LOG (CLOSED LOOP)

Load Download

Per-module datapoints recorded for this run.

Run ID: f1a98533-821d-4a93-b6a2-40abaf5f0334

No evidence loaded yet.

CONFIDENCE

Route confidence 0.00 (0%)

Target resolution 0.00 (0%)

Wetlab (target) 0.00 (0%)

Overall signal 0.00 (0%)

Wet-lab priors are shown for context, not gating.

INTENT & ROUTING

Requested Output	Intent Type	Intent Conf
-CF3	n/a	n/a

Requires Trap
n/a

Mechanism route

- Primary: n/a
- Secondary: n/a
- Mechanisms: n/a

Expert tracks

- None

Success definition

- Any activity: Detectable conversion or substrate turnover under plausible conditions.
- Target specific: Detectable formation of the requested fragment or trapped product under plausible conditions.

MECHANISM CONTRACT

Route ID	Reaction family	Expected nucleophile
n/a	n/a	n/a

Allowed geometries	Policy
n/a	n/a

Mechanism match

- Status: unknown
- Expected: n/a
- Observed: n/a
- Penalty: 0.000

UNITY ARBITRATION

Stage	Recommendation	Penalty
module3	monitor	0.000

Hard veto
false

Weights

- physics: 0.324
- chem: 0.214
- bio: 0.264
- math: 0.198

MODULE 0 PHYSICS AUDIT

Temp K	DeltaG_dagger (kJ/mol)	Eyring k (s^-1)
n/a	n/a	n/a

Horizon (s)	Prior success P
n/a	n/a

Physics-based prior when historical routing data is sparse.

TARGET RESOLUTION

Selection Mode	Canonical Token	Match Count
token	ch__aliphatic	0

Candidate Mode	Scanned Bonds	Candidate Matches
token	4	0

Selected bond

Bond	n/a
Role	n/a
Role Conf	n/a

BOND	ROLE	SCORE	STATUS
None			

MODULE -1 SRE

Status	Bond Type	Primary Role
PASS	ch	ch

Constraint	Confidence	Cache
NONE	0.30	Miss

Attack sites

Electrophile	n/a
Leaving	n/a
Stabilizable	n/a

Eligibility

Serine hydrolase	REJECTED

Route bias

Prefer	p450_oxidation, monooxygenase, radical_transfer
Discourage	serine_hydrolase
Strength	0.20

Module -1 notes

- Warning: fragment build failed: 'Unknown atom_id: f3d0f37e-a6f6-4990-a3d9-f64925da2291_H4'

SRE CPT SIGNALS

CPT	DELTA	DIRECTION	METHOD
None			

Delta uses relative CPT scale; see raw JSON for detail.

STRUCTURE CONTEXT

Atoms	Heavy Atoms	Rings
4	4	0
Hetero Atoms	**Bond Class**	**Bond Type**
3	n/a	n/a

Size proxies

Approx radius	n/a
Approx volume	n/a
Min diameter	n/a

Bond center hint

Atoms	n/a
Anchor	n/a

Functional groups

- halide: 3
- alkyl_halide: 3

COMPUTE & CONSTRAINTS

Scaffolds	TopoGate	QM
n/a	n/a	n/a
MD	**Review**	**Active**
n/a	n/a	n/a
pH Min	**pH Max**	**Temp C**
n/a	n/a	n/a
Metals	**Oxidation**	**Host**
true	true	n/a

Assumptions

- metals_allowed default true
- oxidation_allowed default true

MODULE 1 SUMMARY

Status	Mode	Weights
FAIL	standard	access 0.35, reach 0.45, retention 0.20

Cache Hits	Cache Misses	Cache Size
0	0	0

SCAFFOLD	TOTAL	ACCESS	REACH	RETENTION
None				

Showing top 0 scaffolds.

Physics audit

Temp K	DeltaE model (kJ/mol)	Boltz mean
n/a	n/a	n/a

Boltz min/max	Diffusion cap
n/a	n/a

MODULE 2 SUMMARY

Status	Selected Scaffold	Adjusted Score
DEGRADED_OK	n/a	n/a

Retention Risk	Nucleophile Geo	Best Variant
n/a	n/a	n/a

Mechanism Consistency
n/a

Objective

n/a

Selection rationale

n/a

Retention penalty reasons

- None

Physics audit

DeltaG_dagger base (kJ/mol)	DeltaDeltaG_dagger (kJ/mol)	Strain (kJ/mol)
n/a	n/a	n/a

DeltaG_dagger variant (kJ/mol)	k variant (s^-1)
n/a	n/a

Module 2 warnings

- None

Module 2 errors

- None

VARIANT PROPOSALS

VARIANT	SCORE	MUTATIONS	RATIONALE
None			

MODULE 3 EXPERIMENT PLAN

Status	Batch ID	Plan Score
PASS	d6e1b499-580f-4cda-a8d5-62953f9e7994	0.142

Plan Phys	QC Status	Learning
0.000	N/A	PLANNED

Arms
5

Baseline expected outcome

Baseline activity under user conditions.

Baseline conditions

pH 7.35, T 35.0C

Arms

- A4 • negative_control • no_enzyme • pH 7.35, T 35.0C
- A5 • stress_boundary • Baseline • pH 7.35, T 27.0C
- A1 • baseline • Baseline • pH 7.35, T 35.0C
- A2 • improved_conditions • Baseline • pH 7.35, T 37.0C
- A3 • variant_disambiguation • Baseline vs Baseline • pH 7.35, T 35.0C

Information gain notes

- total_eig=0.825481

Physics audit

Horizon (s)	Expected signal	Noise floor
3600	0.000	0.020

SNR
0.00

MODULE 3 QC

QC Status	Records	Router Updates
N/A	5	0

QC reasons

- None

DIAGNOSTICS

Required next input

- target_bond

Protonation flags

- None

Descriptor status

- None

DATA QUALITY

Warnings

- Token inferred heuristically from input.
- Requested output appears to be a functional group.
- W_MODULE1_VETO: downstream physics/geometry indicates reroute or review.

Errors

- No bonds matched the requested token.